

RESEARCH ARTICLE

ZIF-Derived FeCo Nanoparticles and FeN₄ Single-Atom Sites for High-Performance Oxygen Evolution and Oxygen Reduction Bifunctional Electrocatalysis

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ABSTRACT

Atomically dispersed nitrogen-coordinated iron single atoms on carbon (FeN₄SACs) stand out as highly promising non-precious metal catalysts for the air cathode of zinc-air batteries (ZABs). Herein, we improve bifunctional catalytic performance by uniformly anchoring FeCo nanoparticles onto a 3D cubic FeN₄ substrate through internal vapor-phase etching and surface confinement engineering. In alkaline electrolyte, Fe₁Co₁@FeNC delivers exceptional bifunctionality with an ORR $E_{1/2}$ of 0.904 V and an OER overpotential of 308 mV at 10 mA cm⁻². The catalyst also exhibits excellent long-term cycling stability, with only a negligible $E_{1/2}$ decay of 18 mV after 10,000 consecutive cycles. Notably, Fe₁Co₁@FeNC-based ZABs deliver a peak power density of 167.67 mW cm⁻² and over 600 h of cycling stability, outperforming benchmark Pt/C + RuO₂. Density functional theory (DFT) reveals that the synergy between FeCo nanoparticles and Fe single atoms promotes interfacial electron transfer and downshifts the Fe d-band center. This modulates the electronic configuration and weakens *OH binding, optimizing intermediate adsorption energies to significantly boost the intrinsic bifunctional activity.

1 | Introduction

Since the early 21st century, the global energy crisis has become one of the major bottlenecks restricting sustainable human development. Traditional fossil fuels are nonrenewable energy sources. Their combustion releases massive amounts of CO₂ and toxic pollutants, exacerbating global greenhouse effects and environmental degradation. In this context, the development of efficient, clean, and renewable energy devices, including fuel cells and metal-air batteries, has been extensively promoted [1–3]. Among these technologies, rechargeable zinc-air batteries (ZABs) are considered highly promising next-generation storage systems due to their superior safety, environmental compatibility, and high theoretical energy density (1086 Wh kg⁻¹). Nevertheless, the practical commercialization of ZABs is severely hindered by the sluggish kinetic behavior and limited cycling durability of air cathodes. Both

the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) involve complicated multi-electron and proton transfer processes [4–7]. At present, the platinum-based catalysts (typically commercial Pt/C + RuO₂) serve as the benchmark bifunctional oxygen electrocatalysts. However, high cost, natural scarcity, and poor long-term durability severely restrict their large-scale industrial applications [8–14]. Accordingly, the exploration of high-efficiency non-precious metal catalysts (NPMCs) with excellent bifunctional electrocatalytic activity has become critical [15–19].

Fe-N-C catalysts with well-defined FeN₄ moieties are considered the most competitive alternatives to noble metal catalysts, owing to their high atomic utilization efficiency and tunable electronic structures. Atomically dispersed Fe-N coordination sites have been universally identified as the dominant active centers of Fe-N-C materials for oxygen electrocatalysis [20, 21].

In accordance with the Sabatier principle, optimal catalytic performance requires moderate adsorption strength between catalyst active sites and oxygen-containing intermediates, with binding interactions that are neither excessively strong nor weak [22, 23]. Unfortunately, conventional FeN configurations possess symmetric electronic distribution, which induces deviated adsorption energies toward key oxygen intermediates (e.g., O*, OH*, OOH*) from their optimal values and ultimately deteriorates bifunctional catalytic activity [24–26]. Therefore, precisely modulating the electronic structure of FeN active sites is an effective and imperative strategy to further boost the bifunctional oxygen electrocatalytic performance of Fe-N-C materials.

Previous studies have demonstrated that additional Fe active sites can be constructed by impregnating pre-synthesized Fe-N-C substrates with iron ions followed by high-temperature pyrolysis, and thus effectively enhances catalytic performance [27, 28]. However, excessive iron doping tends to trigger the aggregation and overgrowth of Fe-based nanoparticles during pyrolysis, which impairs intrinsic catalytic activity and reduces active site accessibility.

To date, two mainstream strategies have been developed to modulate the electronic structure of atomic Fe centers and optimize their intrinsic electrocatalytic activity. The first approach involves heteroatom doping, in which partial nitrogen atoms within the FeN skeleton are substituted by foreign heteroatoms (e.g., S, P, B, Se) to construct asymmetric Fe active configurations [29–33]. Heteroatom incorporation can effectively regulate the adsorption and desorption behaviors of oxygen intermediates on Fe active sites. Nevertheless, the random doping nature during high-temperature treatment renders precise structural regulation difficult, greatly limiting its practical application [34, 35].

The second strategy introduces adjacent metal centers (e.g., Pt, Cu, Co) into the Fe-N-C matrix to induce electronic redistribution and generate synergistic catalytic effects [36, 37]. For instance, Mo doping in Fe-N centers can downshift the d-band center, optimize intermediate adsorption/desorption, and form additional MoN active sites to redistribute surface charge and accelerate OH* desorption. Such multisite synergies significantly accelerate electrocatalytic reaction kinetics, endowing Mo-modified Fe-N-C catalysts with outstanding oxygen catalytic activity [38].

More recently, integrating FeN single-atom sites with metal nano-clusters or nanoparticles has emerged as a state-of-the-art strategy to upgrade the catalytic performance of single-atom catalysts (SACs) [39, 40]. Consequently, hybrid catalysts coupling FeN active centers with secondary metal phases have attracted rapidly growing research interest. For example, Zhang et al. reported that integrating iron oxides with Fe single atoms can regulate the charge distribution and crystal field environment of Fe-N sites [41]. Cui et al. verified that the electronic interaction between atomic Fe-N sites and FeC nanoparticles facilitates O₂ adsorption and activation [42]. However, existing studies mainly focus on the modulation of oxygen intermediate adsorption by secondary phases, while the synergistic mechanism toward high-efficiency bifunctional ORR/OER catalysis remains insufficiently explored.

Inspired by the above progress and challenges, this work fabricates a three-dimensional cubic FeCo@FeNC composite catalyst

via a facile two-step calcination strategy using a ZIF-8 precursor. The unique cubic framework effectively alleviates electrode volume expansion/contraction during repeated charge–discharge cycling, guaranteeing exceptional structural stability. Meanwhile, the coupling between FeCo nanoparticles and intrinsic FeN active sites induces strong electronic modulation and favorable interfacial charge transfer, thereby significantly enhancing bifunctional electrocatalytic performance. The optimized Fe₁Co₁@FeNC catalyst delivers a promising ORR half-wave potential (*E*) of 0.904 V and a low OER overpotential of 308 mV at 10 mA cm^{−2}, together with excellent long-term cycling stability. The rational integration of FeNC substrates and FeCo nanoparticles promotes cascade multistep electrocatalytic reactions and favors the complete 4e[−] oxygen redox pathway for efficient intermediate conversion. Density functional theory (DFT) calculations confirm that FeCo nanoparticles promote electron transfer toward FeN active centers, inducing a downward shift of the d-band center. Such electronic modulation weakens the excessive adsorption of OH* intermediates, synchronously boosting both ORR and OER intrinsic activity. Benefiting from the structural and electronic advantages, the assembled ZAB based on the Fe₁Co₁@FeNC air cathode achieves a high peak power density of 167.67 mW cm^{−2} and an ultra-long cycling lifespan of 600 h, which prominently outperforms commercial Pt/C-based benchmark batteries. This work provides a feasible strategy for the rational design and fabrication of high-efficiency bifunctional electrocatalysts and offers new insights for the development of advanced energy conversion and storage systems.

2 | Experimental

Synthesis of catalysts. Using zeolitic imidazolate framework (ZIF) as the precursor template, FeCo alloy nanoparticles were deposited uniformly on the surface of three-dimensional cubic FeN₄ by gas-phase etching and surface confinement, yielding the target FeCo@FeNC catalysts. See the Supporting Information for the specific preparation method.

Catalyst characterization and electrochemical evaluation. A range of materials characterization techniques and instruments was used to characterize the as-prepared catalysts. See the Supporting Information for more electrochemical test results.

3 | Results and Discussion

3.1 | Synthesis and Structural Characterization of Electrocatalysts

The three-dimensional cubic Fe₁Co₁@FeNC catalyst was synthesized by a two-step route, and the detailed synthesis process is shown in Figure 1. Firstly, zinc ions served as metal nodes and 2-methylimidazole as organic ligands; Fe-doped ZIF-8 precursors were obtained by adding cetyltrimethylammonium bromide (CTAB) surfactant and ferric nitrate into aqueous solution, where Fe³⁺ partially replaced Zn²⁺ during crystal growth. A high dosage of 2-methylimidazole could inhibit the hydrolysis of zinc nitrate, ensuring the formation of intact cubic structures [43]. The as-obtained precursors were exfoliated and carbonized at 900 °C

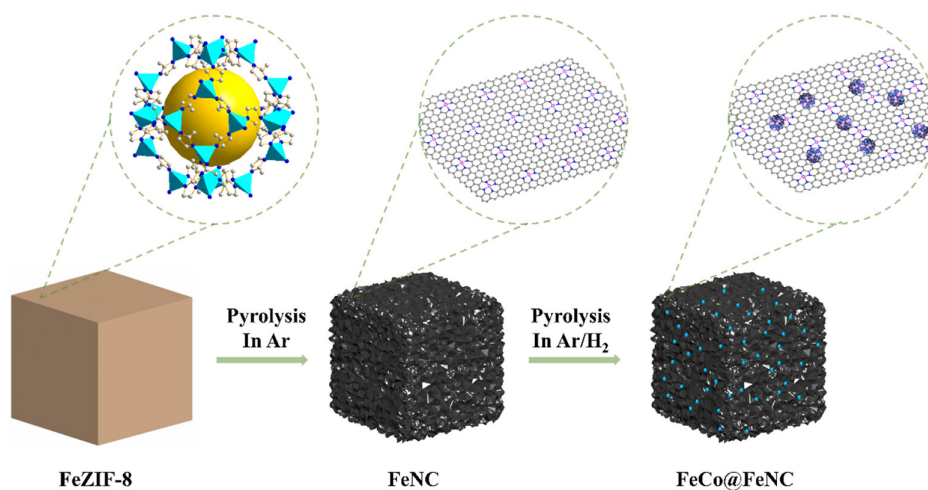


FIGURE 1 | Schematic illustration of the synthesis of FeCo@FeNC catalyst.

to form wrinkled carbon substrates with enlarged specific surface areas. The carbonized products were dispersed in a mixed solution of ferric nitrate and cobalt nitrate, followed by stirring and solvent evaporation to obtain hybrid metal precursors. Finally, the samples were calcined at 950 °C under a reducing mixed atmosphere to grow FeCo nanoparticles. During high-temperature calcination, urea decomposition produced ammonia for chemical etching, forming irregular nanosheets and exposing more active sites after heat treatment [44].

Scanning electron microscopy (SEM), transmission electron microscopy (TEM), and energy-dispersive X-ray spectroscopy (EDS) were adopted to characterize the micromorphology and elemental distribution of FeCo@FeNC. TEM and SEM images reveal that FeCo nanoparticles are uniformly dispersed on the cubic substrate with an average edge length of ~125 nm (Figures S1 and 2a). Further EDS elemental mapping

(Figure 2d–h) demonstrates the homogeneous distribution of Fe, Co, N, and C elements without obvious agglomeration. By adjusting the Fe/Co molar ratio in the impregnation solution of FeNC supports, a series of $\text{Fe}_x\text{Co}_y\text{@FeNC}$ catalysts ($x:y = 1:3-3:1$) were synthesized. All samples present similar cubic morphology to $\text{Fe}_1\text{Co}_1\text{@FeNC}$. However, excessive addition of Fe^{3+} or Co^{3+} leads to decreased size uniformity of metal particles and severe particle agglomeration (Figures S2–S5).

High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images confirm the uniform dispersion of Fe single atoms on carbon supports (Figure 2b). The high-resolution TEM (HRTEM) image and corresponding magnified view (Figure 2c) show a lattice spacing of 2.01 Å for metal nanoparticles, which corresponds to the (110) crystal plane of ordered FeCo alloy. X-ray diffraction (XRD) patterns further verify that the nanoparticles anchored on nanocubes

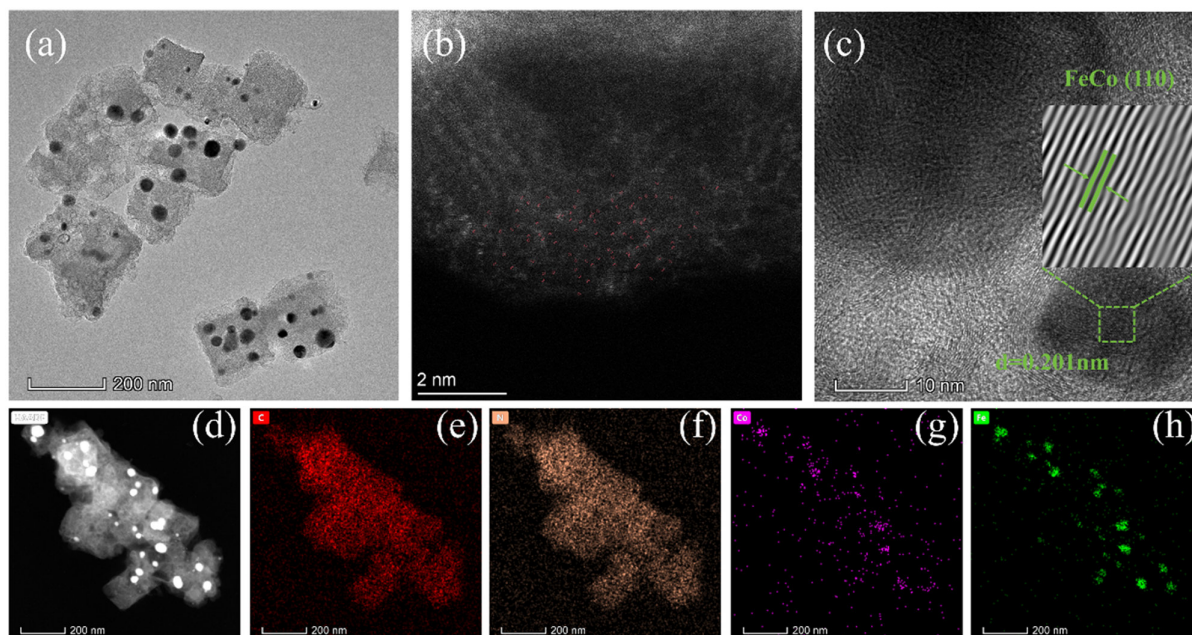


FIGURE 2 | (a) TEM image, (b) HAADF-STEM, (c) HRTEM image of $\text{Fe}_1\text{Co}_1\text{@FeNC}$, and enlarged HRTEM image of the green areas indicated in (d–h) EDS elemental mapping of $\text{Fe}_1\text{Co}_1\text{@FeNC}$.

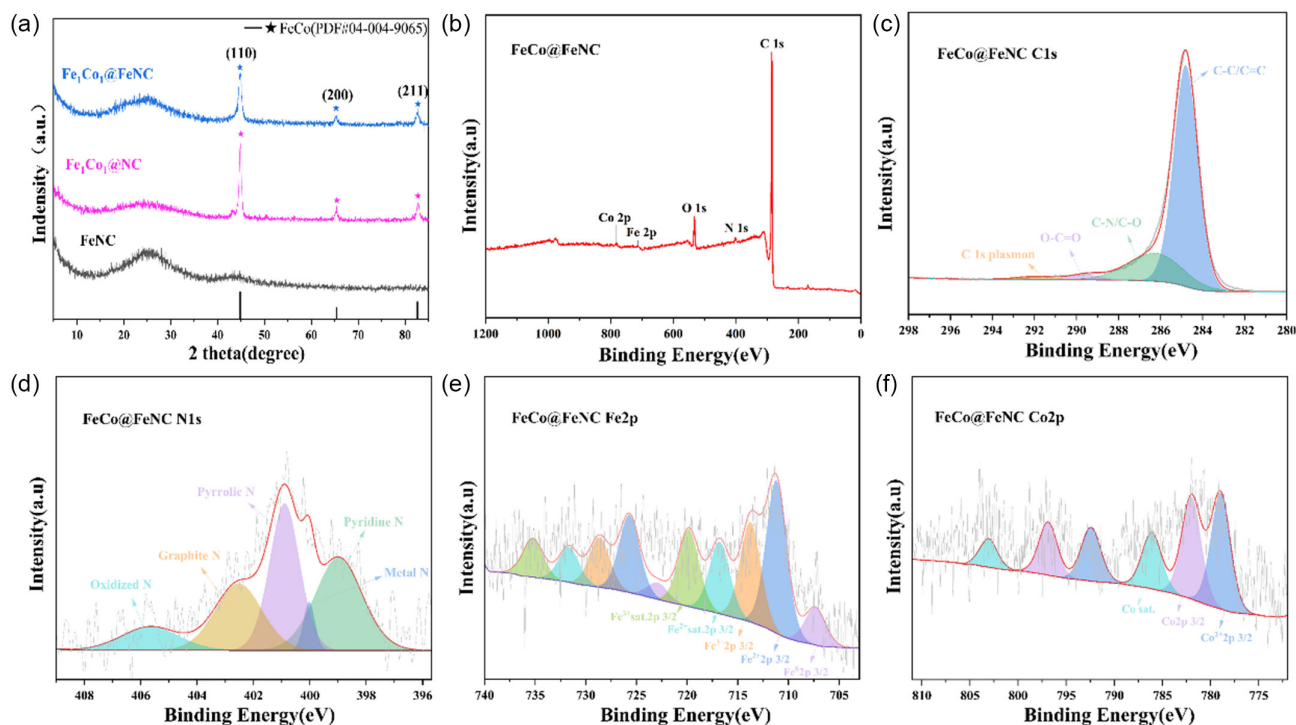


FIGURE 3 | (a) XRD images of Fe₁Co₁@FeNC, Fe₁Co₁@NC, and FeNC; (b) total XPS spectrum of Fe₁Co₁@FeNC; (c) XPS patterns of C 1s; (d) N 1s; (e) Fe 2p (f) Co 2p.

are ordered FeCo alloys. The characteristic diffraction peaks at $\sim 44.8^\circ$, 65.3° and 82.7° are assigned to the (110), (200) and (211) planes of FeCo alloy (PDF#04-004–9065), respectively (Figure 3a).

X-ray photoelectron spectroscopy (XPS) was used to analyze the chemical states, electronic structures and bonding modes of surface elements in FeCo alloy-modified FeNC catalysts (Figure 3b–f). The full-range XPS spectrum (Figure 3b) confirms that the catalyst surface mainly consists of C, N, Fe, and Co elements, which is consistent with the designed composition and verifies the successful synthesis of FeCo@FeNC. The high-resolution C 1s spectrum (Figure 3c) is deconvoluted into four peaks: C–C/C=C (284.8 eV), C–N/C–O (286.2 eV), O–C=O (289.5 eV), and C 1s plasmon peak (291.5 eV). The presence of C–C/C=C bonds indicates the graphitized carbon structure, which facilitates electron transport. C–N bonds prove the successful incorporation of nitrogen into carbon frameworks. Nitrogen doping induces charge rearrangement on adjacent carbon atoms, modulates local electron cloud distribution, and generates active sites favorable for adsorbing ORR intermediates. The N 1s XPS peak (Figure 3d) is deconvoluted into five characteristic peaks: pyridine-N at ~ 398.9 eV, metal-N at ~ 400.0 eV, pyrrole-N at ~ 400.9 eV, graphite-N at ~ 402.5 eV, and oxide-N at ~ 405.6 eV, respectively. The existence of metal-N and pyridine-N hints at the successful preparation of FeN₄. Notably, metal-N moieties and pyridinic-N are widely recognized as highly active sites for oxygen electrocatalysis [45]. The Fe 2p spectrum (Figure 3e) can be fitted into five peaks, including the metals Fe⁰, Fe²⁺, and Fe³⁺ and corresponding satellite peaks. The double peaks at 711.2 and 725.7 eV are attributed to the typical 2p_{3/2} and 2p_{1/2} [46]. For the Co 2p spectrum (Figure 3f), the peaks at 782 and 796.7 eV

belong to metallic Co 2p_{3/2} and Co 2p_{1/2}, the peaks at 778.9 and 792.4 eV are assigned to Co²⁺ 2p_{3/2} and Co²⁺ 2p_{1/2}, while the signals at 786.1 and 803.1 eV are typical satellite peaks of Co species [47]. To explore the coordinate effect between FeCo alloy nanoparticles and FeNC support on electronic structure, we compared the XPS results of FeNC and FeCo@FeNC. The Fe 2p binding energy of FeCo@FeNC shifts negatively relative to pure FeNC, revealing obvious electron transfer between FeCo nanoparticles and FeNC substrates, which effectively regulates the electronic structure of metal active species (Figure S7).

3.2 | Electrochemical Performance

The electrocatalytic performance of the catalysts was systematically evaluated using a conventional three-electrode system, and all potentials were calibrated to the reversible hydrogen electrode (RHE). Linear sweep voltammetry (LSV) curve measurements were conducted in O₂-saturated 0.1 M KOH solution at a rotating speed of 1600 rpm (Figure 4a). The Fe₁Co₁@FeNC catalyst exhibits the optimal ORR activity with a half-wave potential ($E_{1/2}$) of 0.904 V versus RHE, which is superior to Fe₁Co₁@NC (0.863 V), FeNC (0.825 V), NC (0.796 V), and commercial Pt/C (0.833 V). Its limiting diffusion current density (J_L) is 5.36 mA cm⁻², slightly lower than NC (5.62 mA cm⁻²), but higher than Fe₁Co₁@NC (4.85 mA cm⁻²), FeNC (4.92 mA cm⁻²), and Pt/C (4.5 mA cm⁻²). Tafel slope was calculated to evaluate ORR kinetic (Figure 4b). Fe₁Co₁@FeNC exhibits the lowest Tafel slope of 94.5 mV dec⁻¹, much lower than other samples and Pt/C (119.4 mV dec⁻¹), demonstrating that the synergistic effect between FeCo nanoparticles and FeNC greatly accelerates ORR kinetics. We further evaluated the OER catalytic performance of all samples in alkaline media

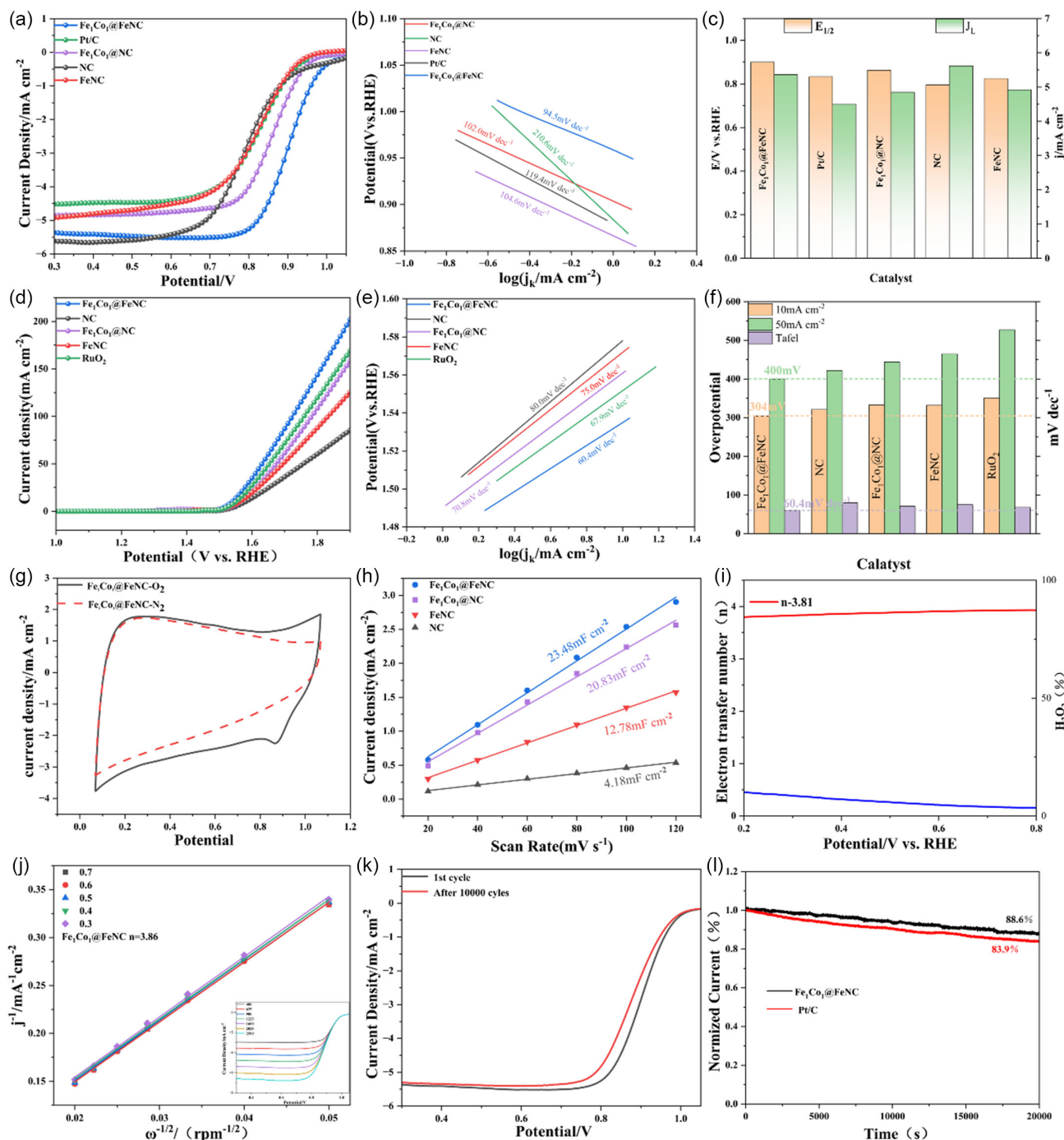


FIGURE 4 | (a) ORR polarization curves (b) Tafel plots corresponding to the ORR (c) Bar chart comparison ($E_{1/2}$ and J_1) (d) OER polarization curves (e) Tafel plots corresponding to the OER (f) Bar chart comparison (η_{10} , η_{50} and Tafel) (g) Cyclic voltammetry (CV) curves of catalysts measured in N₂ and O₂ saturated 0.1 M KOH (h) Cdl plot (i) H₂O₂ yield and electron transfer number (j) Corresponding Koutecky-Levich (K-L) and LSV curves of Fe₁Co₁@FeNC at different rotation rates (k) Aged LSV comparison after 10,000 cycles of CV (l) Timing current response plots for Fe₁Co₁@FeNC.

(Figure 4d). At a current density of 10 mA cm⁻², Fe₁Co₁@FeNC shows an ultra-low overpotential of 308 mV, outperforming all control samples. The overall potential gap (ΔE) between ORR ($E_{1/2}$) and OER overpotential at 10 mA cm⁻² is only 0.63 V, which is competitive against most reported bifunctional electrocatalysts (Table S1). Meanwhile, Fe₁Co₁@FeNC possesses the smallest OER Tafel slope (Figure 4e), indicating accelerated OER reaction kinetics.

Cyclic voltammetry (CV) curves were measured in N₂⁻ and O₂⁻ saturated 0.1 M KOH electrolytes (Figure 4g). No obvious redox peaks are observed for Fe₁Co₁@FeNC in N₂⁻ saturated electrolyte, while a distinct reduction peak appears in the range of 0.7–1.0 V in O₂⁻ saturated electrolyte, confirming its excellent ORR catalytic activity.

The electrochemical double-layer capacitance (C_{dl}) was tested at various scan rates to estimate the electrochemically active

surface area (ECSA). The ECSA is calculated by the formula: $ECSA = C_{dl} \times A_{SA} / C_s$, where A_{SA} is the actual electrode area, and C_s represents the specific capacitance. $Fe_1Co_1@FeNC$ exhibits the highest C_{dl} value of 23.48 mF cm^{-2} (Figure 4h), proving its largest active surface area and highest exposure degree of active sites. Rotating ring-disk electrode (RRDE) measurements were performed to investigate the ORR pathway, H_2O_2 yield, and average electron transfer number (n). As shown in Figure 4i, the electron transfer number of $Fe_1Co_1@FeNC$ remains stable at ~ 3.81 , which is consistent with the value of 3.86 calculated from Koutecky–Levich ($K-L$) plots (Figure 4j). These results are close to the theoretical value of the $4e^-$ ORR pathway. Meanwhile, the catalyst maintains an extremely low H_2O_2 yield over the entire tested potential range, further verifying the dominant $4e^-$ pathway and efficient suppression of H_2O_2 by-products. At the same time, the same electrochemical tests were carried out on $Fe_xCo_y@FeNC$ ($x:y = 1:3 \sim 3:1$). For detailed results, please refer to Supporting Information (S8–S11). Electrochemical impedance spectroscopy (EIS) Nyquist plots (Figure S12) show that $Fe_1Co_1@FeNC$ has the smallest semicircle diameter, corresponding to the lowest charge transfer resistance. Methanol tolerance tests reveal that the LSV curve of $Fe_1Co_1@FeNC$ barely shifts after methanol addition, while commercial Pt/C suffers severe current attenuation (Figure S13). Accelerated durability tests demonstrate outstanding cycling stability of $Fe_1Co_1@FeNC$: after 10,000 consecutive CV cycles, its $E_{1/2}$ only decreases by 18 mV (Figure 4k). Chronoamperometry tests show that the catalyst retains 88.6% of its initial current after 20,000 s of continuous operation, higher than 83.9% of commercial Pt/C (Figure 4l). All results confirm the superior electrochemical stability of $Fe_1Co_1@FeNC$.

3.3 | Oxygen Electrocatalysis Mechanism on $Fe_1Co_1@FeNC$

DFT calculations were carried out to explore the influence of FeCo alloy nanoparticles on the ORR/OER activity of $Fe_1Co_1@FeNC$ and analyze the adsorption behaviors of oxygen-containing intermediates on catalyst surfaces. Calculation details are provided in the Supporting Information. Based on structural characterization results, we established a theoretical model of FeCo nanoparticles anchored on the one-dimensional network structure of FeN_x sites (Figure 5a). A schematic diagram of ORR and OER reaction pathways under alkaline conditions is presented in Figure 5f. The Gibbs free energy diagrams of elementary reaction steps for FeNC and $Fe_1Co_1@FeNC$ were calculated to illustrate the $4e^-$ ORR mechanism (Figure 5b,c), involving $*OOH$, $*O$ and $*OH$ intermediates ($*$ denotes active adsorption sites). At $U=0 \text{ V}$, the total Gibbs free energy decreases continuously, indicating that the ORR is a thermodynamically spontaneous exothermic process. For both FeNC and $Fe_1Co_1@FeNC$, the conversion of $*OH$ to OH^- is identified as the rate-determining step (RDS). At the equilibrium potential of 1.23 V, the Gibbs free energy barrier of the RDS is 1.107 eV for FeNC and 0.806 eV for $Fe_1Co_1@FeNC$. The lower energy barrier suggests that $*OH$ is more easily desorbed from $Fe_1Co_1@FeNC$, and the energy barriers of all elementary steps are closer to 1.23 V, achieving higher reaction efficiency.

Obvious interfacial electron transfer is observed between FeCo nanoparticles and FeNC supports according to the results of Differential charge density and Bader charge analyses (Figure 5e). FeCo nanoparticles transfer a net of 1.41 electrons to the FeNC substrate.

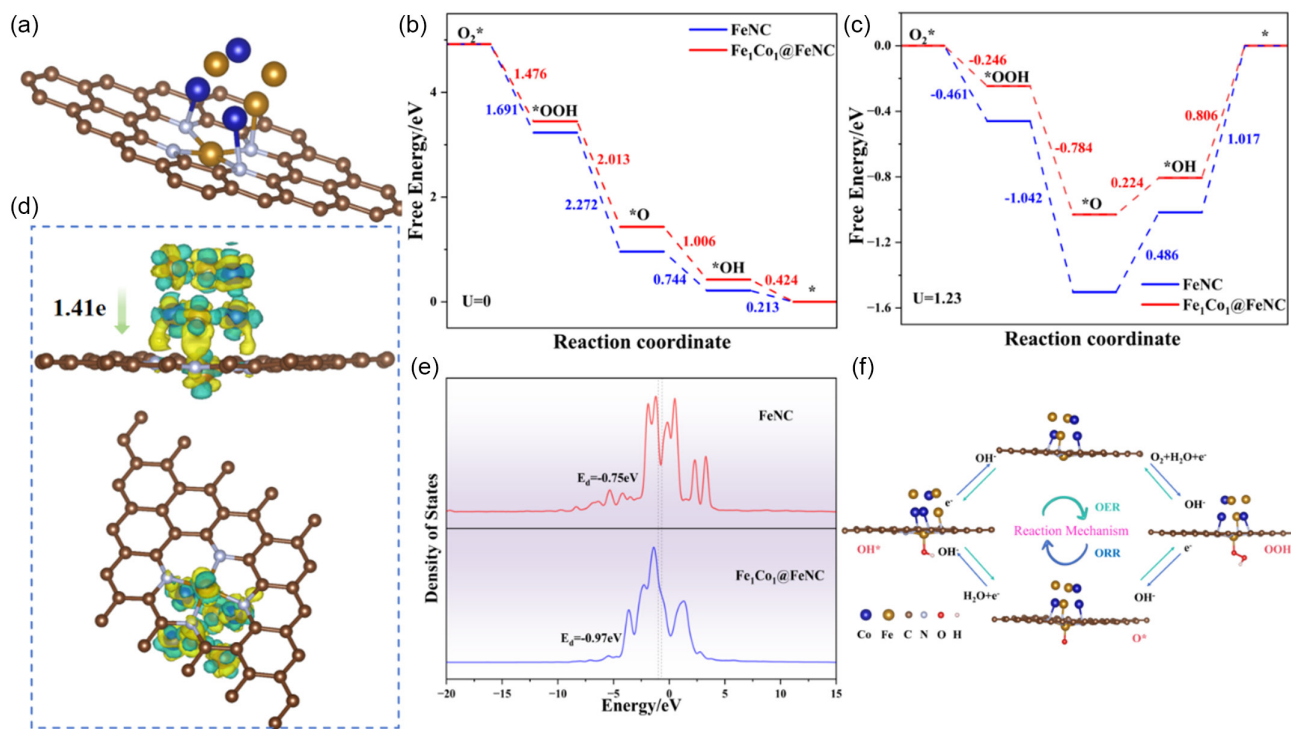


FIGURE 5 | (a) Model diagram of $Fe_1Co_1@FeNC$; (b) ORR process free energy diagrams for $Fe_1Co_1@FeNC$ and FeNC models at pH=14 and equilibrium potential $U=0 \text{ V}$; (c) $U=1.23 \text{ V}$; (d) charge density difference of $Fe_1Co_1@FeNC$; (e) density of states (DOS) of $Fe_1Co_1@FeNC$ and FeNC; (f) schematic illustration of the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER).

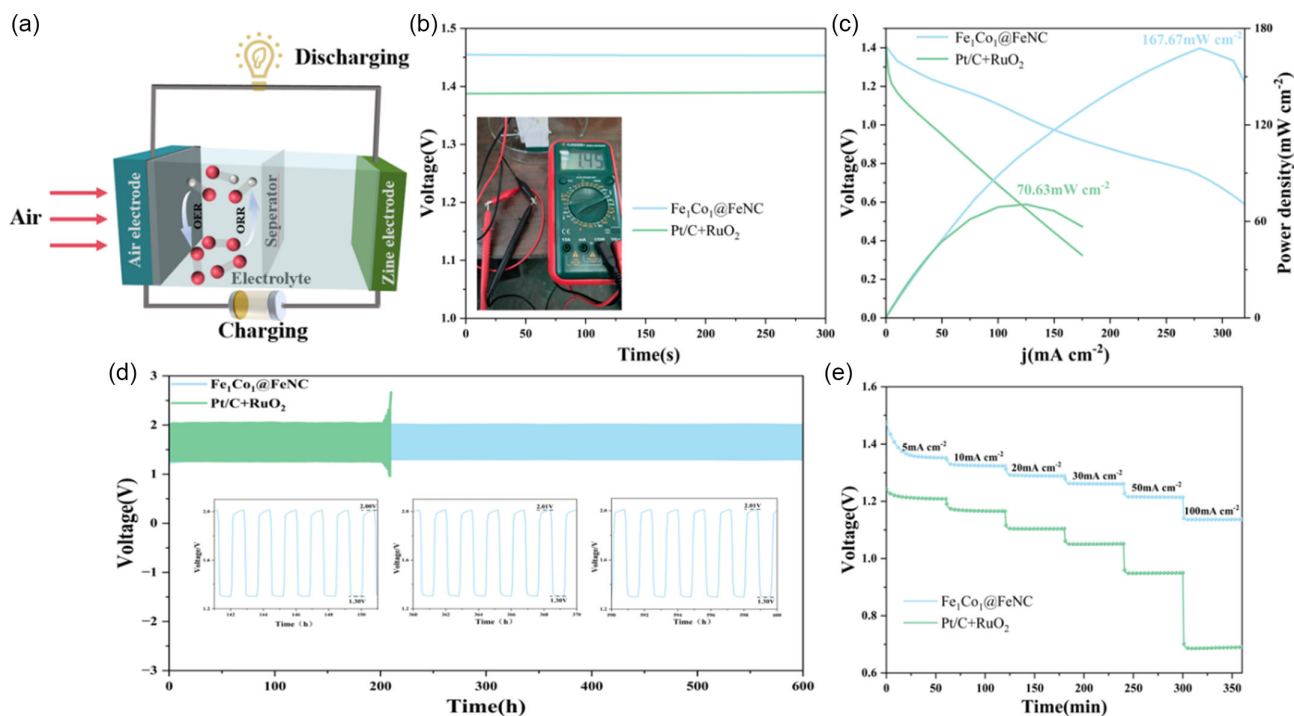


FIGURE 6 | (a) Schematic illustration of the ZAB configuration; (b) Open-circuit voltages; (c) charge–discharge polarization and corresponding power density curves; (d) cyclic stability plot at a charge and discharge current density; (e) discharge plateau at different current densities.

The d-band center is a key descriptor for evaluating the adsorption affinity of reaction intermediates on catalyst surfaces. We calculated the density of states (DOS) and d-band centers of Fe atoms. The d-band center of Fe in pure FeNC is located at -0.75 eV, while it downshifts to -0.97 eV in $\text{Fe}_1\text{Co}_1\text{@FeNC}$. Combined with XPS and charge density results, Fe sites in FeNC act as electron acceptors and capture electrons from FeCo alloys, which modulates the electron filling of Fe d-orbitals and reduces the energy difference between d-orbitals and the Fermi level. The downshifted d-band center weakens the binding strength between Fe sites and oxygen-containing intermediates (*OH and *O), thus accelerating reaction kinetics.

3.4 | Application Evaluation of ZABs

To evaluate the practical performance of the as-prepared catalysts, rechargeable ZABs were assembled using $\text{Fe}_1\text{Co}_1\text{@FeNC}$ as air cathodes. The electrolyte consisted of 6 M KOH and 0.2 M ZnO. For comparison, ZABs with commercial Pt/C + RuO_2 cathodes were fabricated under identical conditions. The ZAB configuration and reaction mechanism are shown in Figure 6a. The ZAB based on $\text{Fe}_1\text{Co}_1\text{@FeNC}$ delivers an open-circuit voltage (OCV) of 1.46 V, higher than the 1.38 V of the Pt/C + RuO_2 -based battery (Figure 6b). The discharge–charge polarization curves and corresponding power density plots (Figure 6c) show that the $\text{Fe}_1\text{Co}_1\text{@FeNC}$ -based ZAB achieves a high peak power density of $167.67 \text{ mW cm}^{-2}$, far exceeding 70.63 mW cm^{-2} of the reference battery. Galvanostatic charge–discharge cycling tests were conducted at 4 mA cm^{-2} to assess long-term stability (Figure 6d). The $\text{Fe}_1\text{Co}_1\text{@FeNC}$ -based ZAB maintains stable charge–discharge performance over 600 h, which is much superior to the Pt/C + RuO_2 counterpart. After long-term cycling, the voltage gap between discharge and charge

is only 0.71 V, demonstrating excellent cycling stability. The robust performance is attributed to the unique cubic structure, which possesses high mechanical strength and effectively relieves volume expansion/contraction of electrodes during cycling, avoiding structural collapse of catalysts. Two series-connected $\text{Fe}_1\text{Co}_1\text{@FeNC}$ -based ZABs can successfully light up an LED panel (Figure S16). The ZAB with $\text{Fe}_1\text{Co}_1\text{@FeNC}$ exhibits a discharge specific capacity of $842.11 \text{ mAh g}^{-1}$, higher than $711.84 \text{ mAh g}^{-1}$ of Pt/C (Figure S17), accompanied by a more stable discharge plateau. Discharge curves at various current densities further verify the stable voltage output of $\text{Fe}_1\text{Co}_1\text{@FeNC}$ (Figure 6e). All results demonstrate that $\text{Fe}_1\text{Co}_1\text{@FeNC}$ is a promising bifunctional electrocatalyst for high-power and long-lifespan rechargeable ZABs.

4 | Conclusions

In summary, we successfully synthesized three-dimensional cubic FeCo@FeNC composite catalysts via a two-step calcination method using ZIF precursors, which integrate FeCo alloy nanoparticles and FeNC single-atom sites and realize strong synergistic effects between atomic and nanoscale metal components. DFT calculations verify that FeCo nanoalloys effectively regulate the electronic structure of FeNC substrates and downshift the d-band center of Fe active sites, which weakens the binding energy between Fe sites and $\text{*OH}/\text{*O}$ intermediates. The optimized $\text{Fe}_1\text{Co}_1\text{@FeNC}$ catalyst exhibits excellent bifunctional oxygen electrocatalytic performance. It delivers an ORR half-wave potential of 0.904 V and an OER overpotential of 308 mV at 10 mA cm^{-2} . Meanwhile, the catalyst possesses remarkable stability; the $E_{1/2}$ only decays by 18 mV after 10,000 CV cycles, and 88.6% of the initial current is retained

after 20,000 s of chronoamperometry. This work provides a novel strategy to modulate the electronic structure of SACs via nanoalloy modification and promotes the development of high-performance single-atom electrocatalysts.

Author Contributions

Lin Li: writing – review & editing, writing – original draft, visualization, methodology, investigation, conceptualization. **Lizi He:** resources, funding acquisition. **Ning Han:** supervision, methodology, conceptualization. **Xiaoyu Li:** resources. **Yi Hu:** methodology, supervision. **Cong Han:** resources, software.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.